

(Mino) Mutian XU

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Education

- 2021–2025 : **Ph.D. in Computer Information and Engineering**, The Chinese University of Hong Kong, Shenzhen, **Representative of Presidential Award for Outstanding Doctoral Students**.
- 2018–2020 : **M.Sc. in Computer Engineering (GPA: 3.592/4.0)**, University of California, Irvine.
- 2014–2018 : **B.Eng. in Radio Wave Propagation & Antenna (GPA: 3.56/4.0)**, Xidian University, Xi'an.

Work Experience

- Sep, 2025 – **Multimedia Laboratory, Nanyang Technological University, Singapore**, Research Fellow.
Now *Embodied AI*
- Advisor : **Prof. Ziwei Liu**
- June, 2020 – **CVMI Lab, The University of Hong Kong, Hong Kong SAR**, Research Assistant.
Feb, 2021 *3D Point Cloud Convolution*
- Advisor : **Prof. Xiaojuan Qi**
- July, 2019 – **Multimedia Lab, Shenzhen Institutes of Advanced Technology, Shenzhen**, Visiting Student.
Nov, 2019 *3D Object Point Cloud Classification and Segmentation*
- Advisor : **Prof. Yu Qiao** and **Dr. Zhipeng Zhou**

Publications (* denotes equal contribution, † indicates project lead, ‡ means corresponding author)

REFEREED CONFERENCE PUBLICATIONS

Xianggang Yu *, **Mutian Xu** †*, Yidan Zhang *, Haolin Liu *, Chongjie Ye *, *et al.*, and Xiaoguang Han. MVImgNet: A Large-scale Dataset of Multi-view Images. In *CVPR 2023*, **200+ citations**.

Mutian Xu *, Runyu Ding *, Hengshuang Zhao, and Xiaojuan Qi. PAConv: Position Adaptive Convolution with Dynamic Kernel Assembling on Point Clouds. In *CVPR 2021*, **700+ citations**.

Mutian Xu, Chongjie Ye, Haolin Liu, Yushuang Wu, Jiahao Chang, and Xiaoguang Han. Stable-Sim2Real: Exploring Simulation of Real-Captured 3D Data with Two-Stage Depth Diffusion. In *ICCV 2025*, **Highlight (2.4%)**.

Mutian Xu *, Pei Chen *, Haolin Liu, and Xiaoguang Han. TO-Scene: A Large-scale Dataset

for Understanding 3D Tabletop Scenes. In *ECCV 2022*, **Oral Presentation (2.7%)**.

Mutian Xu, Xingyilang Yin, Lingteng Qiu, Yang Liu, Xin Tong, and Xiaoguang Han. SAMPro3D: Locating SAM Prompts in 3D for Zero-Shot Instance Segmentation. In *3DV 2025*, **100+ GitHub stars**.

Mutian Xu *, Junhao Zhang *, Zhipeng Zhou, Mingye Xu, Xiaojuan Qi, and Yu Qiao. Learning Geometry-Disentangled Representation for Complementary Understanding of 3D Object Point Cloud. In *AAAI 2021*, **200+ citations**.

Yiming Hao *, **Mutian Xu ***, Chongjie Ye, Jie Qin, Shunlin Lu, Yipeng Qin, and Xiaoguang Han. LoFA: Learning to Predict Personalized Priors for Fast Adaptation of Visual Generative Models. In *CVPR 2026*.

Mingye Xu *, **Mutian Xu ***, Tong He, Wanli Ouyang, Yali Wang, Xiaoguang Han, and Yu Qiao. MM-3DScene: 3D Scene Understanding by Customizing Masked Modeling with Informative-Preserved Reconstruction and Self-Distilled Consistency. In *CVPR 2023*.

David Junhao Zhang, **Mutian Xu †**, Jay Zhangjie Wu, Chuhui Xue, Wenqing Zhang, Xiaoguang Han, Song Bai, and Mike Zheng Shou. Free-ATM: Harnessing Free Attention Masks for Representation Learning on Diffusion-Generated Images. In *ECCV 2024*.

Hongxiang Zhao *, Xingcheng Liu *, **Mutian Xu †**, Yiming Hao, Weikai Chen, and Xiaoguang Han. TASTE-Rob: Advancing Video Generation of Task-Oriented Hand-Object Interaction for Generalizable Robotic Manipulation. In *CVPR 2025*.

Lingteng Qiu *, Guanying Chen *, Xiaodong Gu *, Qi Zuo, **Mutian Xu**, Yushuang Wu, *et al.*, and Xiaoguang Han. RichDreamer: A Generalizable Normal-Depth Diffusion Model for Detail Richness in Text-to-3D. In *CVPR 2024*, **Highlight (2.8%), 200+ citations**.

Lingteng Qiu *, Guanying Chen *, Jiapeng Zhou, **Mutian Xu**, Junle Wang, and Xiaoguang Han. REC-MV: Reconstructing 3D Dynamic Cloth from Monocular Videos. In *CVPR 2023*.

Under Submission

Yidan Zhang, **Mutian Xu ‡**, Yiming Hao, Kun Zhou, Jiahao Chang, Xiaoqiang Liu, Pengfei Wang, Hongbo Fu and Xiaoguang Han. VC-Agent: An Interactive Agent for Customized Video Dataset Collection.

Xingyilang Yin, Jiale Wang, Xi Yang, **Mutian Xu**, Xu Gu and Nannan Wang. Unleashing the Multi-View Fusion Potential: Noise Correction in VLM for Open-Vocabulary 3D Scene Understanding.

REFEREED JOURNAL ARTICLES

Chaoqi Chen, Yushuang Wu, Qiyuan Dai, Hong-Yu Zhou, **Mutian Xu**, Sibe Yang, Xiaoguang Han, and Yizhou Yu. A Survey on Graph Neural Networks and Graph Transformers in Computer Vision: A Task-Oriented Perspective. *TPAMI 2024*, **200+ citations**.

Awards & Certificates

- 2025 **Presidential Award for Outstanding Doctoral Students of CUHKSZ**
- 2025 **Outstanding Reviewer, ICCV 2025 (2.7%)**
- 2025 **Outstanding Reviewer, CVPR 2025 (5.6%)**
- 2024 **WAIC Youth Outstanding Paper Nomination Award** (世界人工智能大会青年优秀论文提名奖, MVLingNet), 2024
- 2023 **Top Reviewer, NeurIPS 2023 (9.9%)**
- 2023 **Outstanding Reviewer, CVPR 2023 (3.3%)**
- 2023 **CCF Excellent Graphics Open-Source Dataset** (中国计算机学会年度优秀图形开源数据集奖, MVLingNet), 2023
- 2022/24 Outstanding Teaching Assistant Award of CUHKSZ

Professional Services

- Journal Reviewer TIP, IJCV, TVCG, NEUCOM, TMM, TMC, MVAP
- Conference Reviewer CVPR, ICCV, ECCV, SIGGRAPH Asia, ICLR, ICML, NeurIPS, 3DV, IJCAI, WACV, ACCV

Talks

- 2024 “学术新人修炼手册 (*Why Do I Need Papers*) ? ” , VALSE Webinar 2024
- 2023 **Outstanding Student Forum** of VALSE 2023
- 2023 **Outstanding Student Forum** of China 3DV 2023
- 2023 Youth PhD Talk - ECCV 2022, invited by AI-TIME

Teaching Assistantship

- CSC1001: Introduction to Computer Science: Programming Methodology**, CUHKSZ, *Leading TA*.
- CSC1002: Computational Laboratory**, CUHKSZ.
- CSC3002: Introduction to Computer Science: Programming Paradigms**, CUHKSZ.

Previous Mentorship

- 2022 Pei Chen, M.Phil. at CUHKSZ, now at ByteDance
- 2024 Xingyilang Yin, B.Eng. at Xidian University, now Ph.D. student at Great Bay University

Miscellaneous

- 3rd Place in the 31st School Singer Contest, Xidian University
- Piano Professional Certificate Level 10